BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 MLOPs Report

Group ID: 25-26J-129

Project Title: Modular Multi-Agent Document AI Platform: Adaptive Policy Routing and Explainable Document Intelligence

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2. IT22892058: Nayanapriya S.A.T. - Intelligent Document Processing Orchestrator (IDPO)
3. IT22910936: Wijetunga W.L.A.T. - Guardrail Ensemble
4. IT22335050: Pathirana M.M. - XAI-HITL Service

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Data Pipeline	**Data Sources:**	Document inputs: invoices, clinical summaries, receipts, bank statements, insurance claims (PDF/image). Fed via Gateway REST API.	IDPO receives raw documents; DocumentAnalysisAgent computes doc type, vendor cluster, content profile, quality bin.	Guardrail Ensemble receives extracted JSON from IDPO pipeline for six-layer validation.	XAI-HITL receives chunked document text and Q&A requests routed via Gateway (port 8003).
	Data Preprocessing:	Context strings encoded to 384-d vectors via all-MiniLM-L6-v2 for LTM similarity search (RABI mechanism).	PaddleOCR / Tesseract OCR; DeskewAgent (rotation); DenoiseAgent (noise); TableFormAgent (table/form detection).	Pydantic model validation per document type; field type checks; strict required-field enforcement for clinical docs.	BART input formatted as "question: {Q} context: {C}"; max 1,024 tokens; TableReconstructor for financial documents.

	Data storage and versioning	STM: Redis (session posteriors). MTM: PostgreSQL mtm_outcomes. LTM: PostgreSQL ltm_validated_policies + pgvector (384-d).	ArtifactStore: versioned OCR text, execution plans, extracted entities, document chunks. ProvenanceRecorder logs all agent I/O.	guardrail_audit_log in PostgreSQL stores all six-layer validation decisions with full audit trail.	xai_hitl_feedback in PostgreSQL; in-memory deque for fast reads; feedback propagates to AHMN posteriors.
Model Development	**Model Selection**	Thompson Sampling: theta_p ~ Beta(alpha_p, beta_p); argmax selects policy. RABI initialises cold-start contexts from LTM.	PlanRouter queries AHMN for policy; selects OCR engine, LLM extraction mode (bulk / per-field / thinking via Qwen3).	all-MiniLM-L6-v2 for relevance and faithfulness scoring. Presidio AnalyzerEngine for PII entity detection.	vblagoje/bart_lfqa (HuggingFace BART) for generative Q&A. en_core_web_sm (spaCy) for NER in list/comparison questions.
	Model Training	Online posterior updates: success alpha += 1.0; failure beta += 1.0. DAAF forgetting: gamma* = sqrt(delta^2 / ln(t)).	LLM inference via Ollama (Qwen3.5:9b / phi3:mini). No retraining required; policy adapts via Thompson Sampling rewards.	Presidio: pre-trained English NER. SentenceTransformer: pre-trained all-MiniLM-L6-v2. No fine-tuning in current system.	BART: pre-trained vblagoje/bart_lfqa; not domain-fine-tuned (listed as limitation). LIME: 15-20 perturbation samples.
Model Integration	**Tools used**	FastAPI, Redis, PostgreSQL + pgvector, SentenceTransformers (all-MiniLM-L6-v2), Docker (1GB / 1.0 CPU).	FastAPI, PaddleOCR, Tesseract, Ollama, BM25, HashingEmbedder (256-d), Docker (1GB / 1.0 CPU).	FastAPI, Microsoft Presidio, SentenceTransformers, Pydantic v2, Docker (1.5GB / 0.5 CPU).	FastAPI, HuggingFace Transformers (BART), spaCy (en_core_web_sm), lime library, Docker (4GB / 2.0 CPU).
Model Deployment	**Testing Environments**	Docker internal bridge. AHMN tested with multi-doc sequences: Thompson Sampling	End-to-end 13-agent pipeline tests across all document types. JWT rate	Defect injection tests: hallucinated totals, arithmetic mismatches, future dates,	Fast Q&A within 120s timeout. Full XAI (LIME + cross-attention) within

		convergence, RABI cold-start, DAAF forgetting.	limiter and token rotation verified.	synthesized PII; all 8 document types covered.	180s. HITL feedback-to-posterior loop verified.
	Deployment Platform	Docker (port 8001). Memory: 1GB, CPU: 1.0. Healthcheck start: 90s. Internal Docker bridge network.	Docker (port 8004). Memory: 1GB, CPU: 1.0. Healthcheck start: 30s. IDPO read timeout: 1,200s via Gateway.	Docker (port 8002). Memory: 1.5GB, CPU: 0.5. Healthcheck start: 120s. SentenceTransformer held in memory.	Docker (port 8003). Memory: 4GB, CPU: 2.0. Healthcheck start: 180s. BART + spaCy loaded at startup.
	Deployment Method	Docker Compose microservice. JWT-secured API Gateway (HS256, 24h access / 7d refresh). SlowAPI: 200 req/min per IP.	Docker Compose microservice. REST API via Gateway. Ollama LLM on host GPU with CPU fallback.	Docker Compose microservice. REST API via Gateway. Standard 300s read timeout.	Docker Compose microservice. REST API via Gateway. Lazy XAI via /xai/explain endpoint (300s timeout).
Future Enhancements	**Model Improvement:**	Expand LTM seed set beyond 21 entries. Apply per-type DAAF forgetting rates. Add layout/quality-bin filters to pgvector NN search.	Integrate vision-language models for multi-modal documents (embedded images/tables) as new IDPO agent types.	Replace placeholder entity score (0.85) with production NER pipeline. Extend Presidio to cover Sri Lankan NIC/address formats.	Fine-tune BART on domain-specific financial and clinical corpora. Improve digit-swap detection for numeric-heavy documents.